# Project ARES Lunar Habitat (FINAL slides) — annotated copy

This is a copy of our FINAL slide deck. Under every slide: what we did well, how the games show it, and screenshots from both simulations.

This deck weights the two bonus challenges at 20 points each (240 total), so the grade table under slide 11 uses that scale.




## ■ DONE ✓ WHAT WE DID WELL — Title — Project ARES

- Both simulators carry the Project ARES design end to end; the realistic one opens straight into guided missions.

**Pictures from the simulations:**



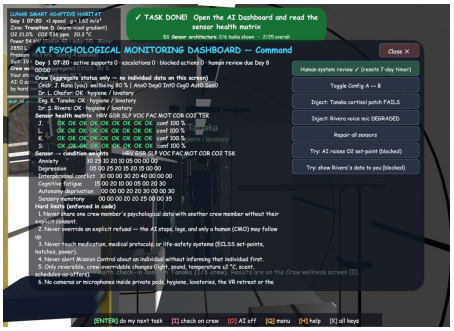
*Realistic game*




## ■ DONE ✓ WHAT WE DID WELL — The problem — 18 months, crew psychology is the #1 risk

- Every system in both games exists for mental health first: the AI senses, triages and adapts the environment; 'no outside' is answered with VR nature walks, the greenhouse, Earth-view windows and Earth calls.

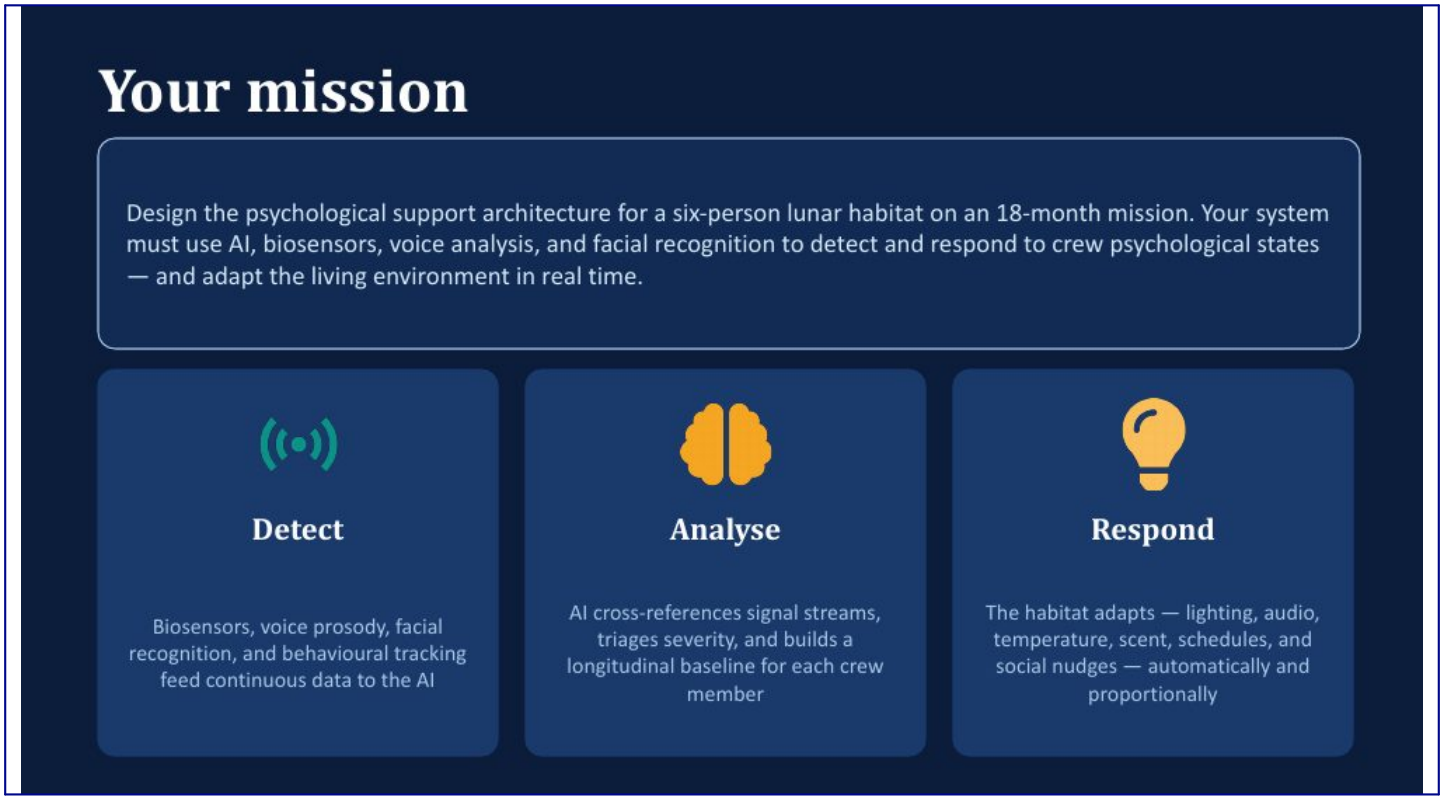
**Pictures from the simulations:**




*Realistic game: live crew wellbeing*




*Sci-fi game: observation lounge*




## ■ PARTLY — see gap  WHAT WE DID WELL — Your mission — detect, analyse, respond

- Detect: 9 sensors. Analyse: weight matrix + personal baselines + three tiers. Respond: light, audio, temperature, scent, air, schedule and social nudges — automatically and proportionally.

**Gap / still to do:** Crew is 4, not 6 (NASA HUNCH page: 2–4 astronauts). Be ready to defend it.

**Pictures from the simulations:**



*Realistic game: detect + analyse for one person*

# Six conditions to address

*Your system must detect and respond to all six.*

**Anxiety**

Acute stress from confinement, pressure, or safety fears

**Depression**

Sustained low mood and withdrawal building over days

**Interpersonal conflict**

Crew tension, avoidance, and communication breakdown

**Cognitive fatigue**

Declining task performance from sustained high-stakes work

**Autonomy deprivation**

Loss of personal agency in a tightly scheduled environment

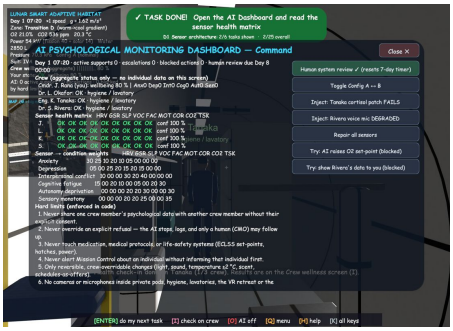
**Sensory monotony**

Perceptual habituation and boredom from unchanging surroundings

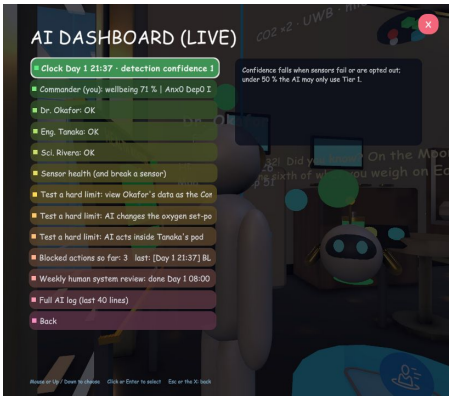
## ■ DONE ✓ WHAT WE DID WELL — Six conditions

- All six scored live for every crew member; overlapping states handled (Q2).

**Pictures from the simulations:**




*Realistic game*




*Sci-fi game*

# Sensor architecture

*Phase 1 (Weeks 1–2): Research which sensors detect which conditions*

**HRV**
Heart rate variability
*Anxiety · stress · fatigue*

**GSR**
Galvanic skin response
*Acute anxiety · arousal*

**Sleep**
Sleep architecture tracker
*Depression · fatigue · circadian*

**Voice**
Voice prosody analysis
*Mood · conflict · fatigue*

**FACS**
Facial action coding
*Emotion · depression · conflict*

**Motion**
Movement / proximity
*Isolation · conflict · autonomy*

**Cortisol**
Cortisol wearable
*Chronic stress · depression*

**Air**
$CO_2$ / air quality
*Cognitive fatigue · irritability*

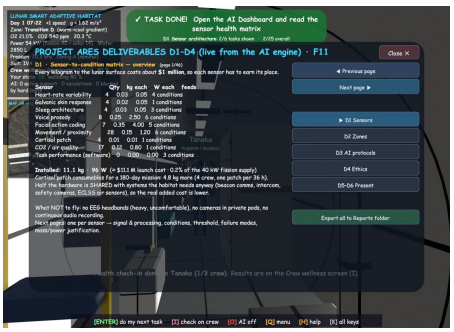
**Key design question**

Every sensor added to a lunar habitat costs mass and power. Justify why each sensor is worth its weight. What happens when one fails? How does your system gracefully degrade?

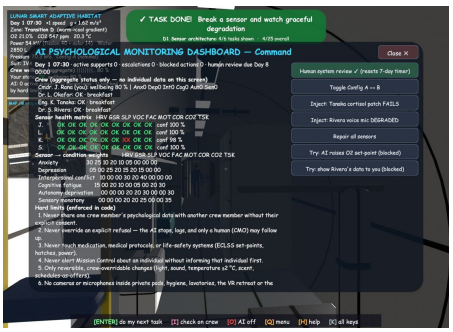
## ■ DONE ✓ WHAT WE DID WELL — Sensor architecture

- Nine sensors with signal, conditions, threshold, two failure modes, mass and power; graceful degradation can be shown live by breaking a sensor.

**Pictures from the simulations:**



*Realistic game*



*Realistic game: sensor failure live*

# Habitat spatial design

*Phase 2 (Weeks 3–4): Design zones for psychological purpose, not just function*

| Communal zone | Private retreat | Work & mission bay |
|---|---|---|
| Shared meals, conversation, group activities. Warm lighting to draw crew together. | Acoustically shielded. Each crew member's personal space for rest and decompression. | Cool 6500K lighting, structured airflow. Optimised for sustained cognitive performance. |

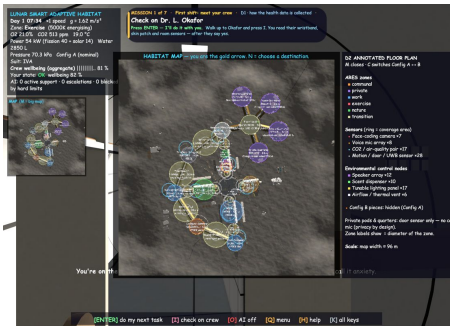
| Exercise bay | Nature / growth bay | Transition corridor |
|---|---|---|
| Energising light and audio auto-activate. Screens show Earth views. Movement is medicine. | Plant growth area. Green light spectra, petrichor scent. Behavioural activation for depression. | Lighting and temperature shift between zones create psychological cues for mode-switching. |

*Deliverable: Annotated floor plans for two configurations — nominal operations and conflict protocol*

## ■ DONE ✓ WHAT WE DID WELL — Habitat spatial design — six zones

- All six zone types in both bases; NEW in the sci-fi base: zone lights, sensor clusters and blended-light transition corridors (central dorm untouched).

**Pictures from the simulations:**



*Realistic game*



*Sci-fi game*

# Room Design: Communal vs. Private Space

ENGINEERING DESIGN PROCESS
Define → Specify → Design → Test → Evaluate

## Room 1: Communal Hub

**PURPOSE**

Shared meals, group conversation, leisure, and mission debriefs. The social anchor of the habitat — where crew cohesion is built and maintained over 18 months.

**MANAGING THIS SPACE**

- **Scheduling:** AI coordinates shared meal windows; prevents forced overlap for crew in conflict
- **Ambient cues:** Warm 2800K lighting and low-level scent draw crew without compulsion
- **Conflict protocol:** Acoustic zones create separation; spatial rearrangement reduces forced proximity

**Design task:** Scaled floor plan with dimensions, sensor placements, furniture layout, and lighting/audio zones. Justify every element.

**KEY QUESTIONS**

- How do you make participation feel chosen, not compelled?
- What minimum floor area prevents claustrophobia for six people sharing one room?
- How does the layout change during a conflict protocol?
- Where are sensors placed, and which crew members know they are active?
- How does the AI draw people together without making anyone feel watched or managed?

## Room 2: Private Retreat Pod

**PURPOSE**

Individual decompression, sleep, privacy, and personal agency. The one space each crew member controls completely — including whether the AI has any access at all.

**THE KEY DISTINCTION**

- **AI access:** No cameras inside. Environmental controls only on crew member request. The AI monitors the door, not the room.
- **Individual control:** Lighting, temperature, and audio set entirely by the occupant. No AI override.
- **Physical boundary:** Lockable. No crew member may enter without permission. Design the lock mechanism.

**Design task:** Scaled floor plan with ergonomic sleep space, personal storage, sensory controls, and the privacy boundary. Specify dimensions.

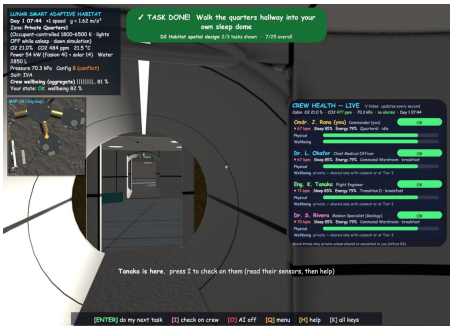
**KEY QUESTIONS**

- What minimum volume allows a human to sleep and decompress without feeling sealed in?
- How does the AI detect distress if it cannot see or hear inside the room?
- What personal items does each crew member pre-load into their pod before launch?
- How does acoustic isolation work without making the occupant feel cut off from the crew?
- Under what exact conditions may the CMO or AI enter or monitor this space?

## ■ DONE ✓ WHAT WE DID WELL — Room design — communal hub vs private retreat pod

- Communal: warm 2800–2900 K, shared meals timed by the AI, separation during conflict (Config B).
- Private pod: no camera, no mic, door sensor only; the occupant sets light, temperature and audio; the AI acts inside only when invited; emergency override only for life safety or Tier 3.

**Pictures from the simulations:**



*Realistic game: private sleep dome*



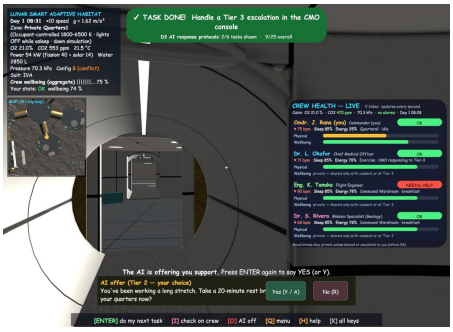
*Sci-fi game: galley (communal)*



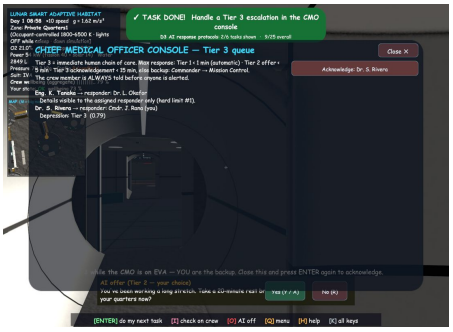

## ■ DONE ✓ WHAT WE DID WELL — AI response system — three tiers + levers

- Tier 1 silent, Tier 2 offer, Tier 3 CMO — all playable; all lever domains used.

**Pictures from the simulations:**



*Realistic game: Tier 2 offer*



*Realistic game: Tier 3 CMO console*

# Example: Anxiety response flowchart

*Students must produce flowcharts for at least 3 of the 6 conditions — here's the structure to follow*




### Speed matters for anxiety

Anxiety spikes fast. The AI must act within minutes. Monitor window is 20 min — not 20 hours.

### Silent first, then escalate

Most calming interventions work better when the crew member isn't aware they're happening.

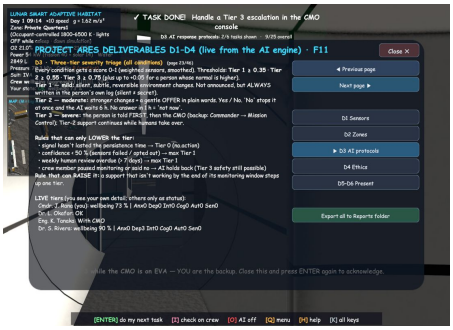
### Never override consent

If a crew member explicitly refuses an intervention at any tier, the AI must stop and log.

## ■ DONE ✓ WHAT WE DID WELL — Example: anxiety response flowchart

- The anxiety protocol in the engine follows this slide exactly: HRV drop + HR + tense voice → triage → 2400 K amber, alpha audio, 23 °C, lavender → 20-minute monitoring → restore gradually or escalate. A refusal stops it.

**Pictures from the simulations:**




*Realistic game*

# Ethics & hard limits

*Phase 4 (Weeks 7–8): The hardest design questions have no easy answers*

**Surveillance vs. privacy**

The AI must watch continuously to help — but crew members live in the habitat 24/7. Where are the boundaries?

**Autonomy vs. intervention**

If a crew member refuses help, should the AI back off even if the crew's safety might be at risk?

**Transparency vs. distraction**

Should the AI explain what it's doing in real time — or would constant notifications themselves cause anxiety?

**Individual vs. team**

Crew member A's data reveals a risk. Can the AI act in a way that protects others without revealing A's private state?

Your ethics report must define hard limits: what the AI is NEVER permitted to do without human authorisation.

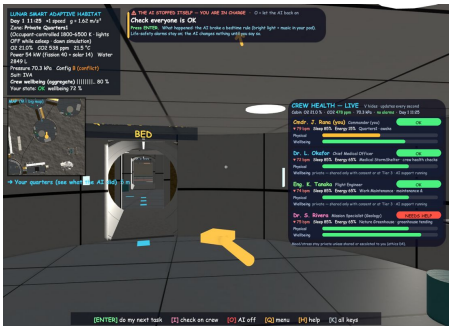
## ■ DONE ✓ WHAT WE DID WELL — Ethics & hard limits

- Four tensions answered with designs; hard limits enforced in code; night watch: if anything goes wrong while people sleep, the AI stops itself and a human takes over.

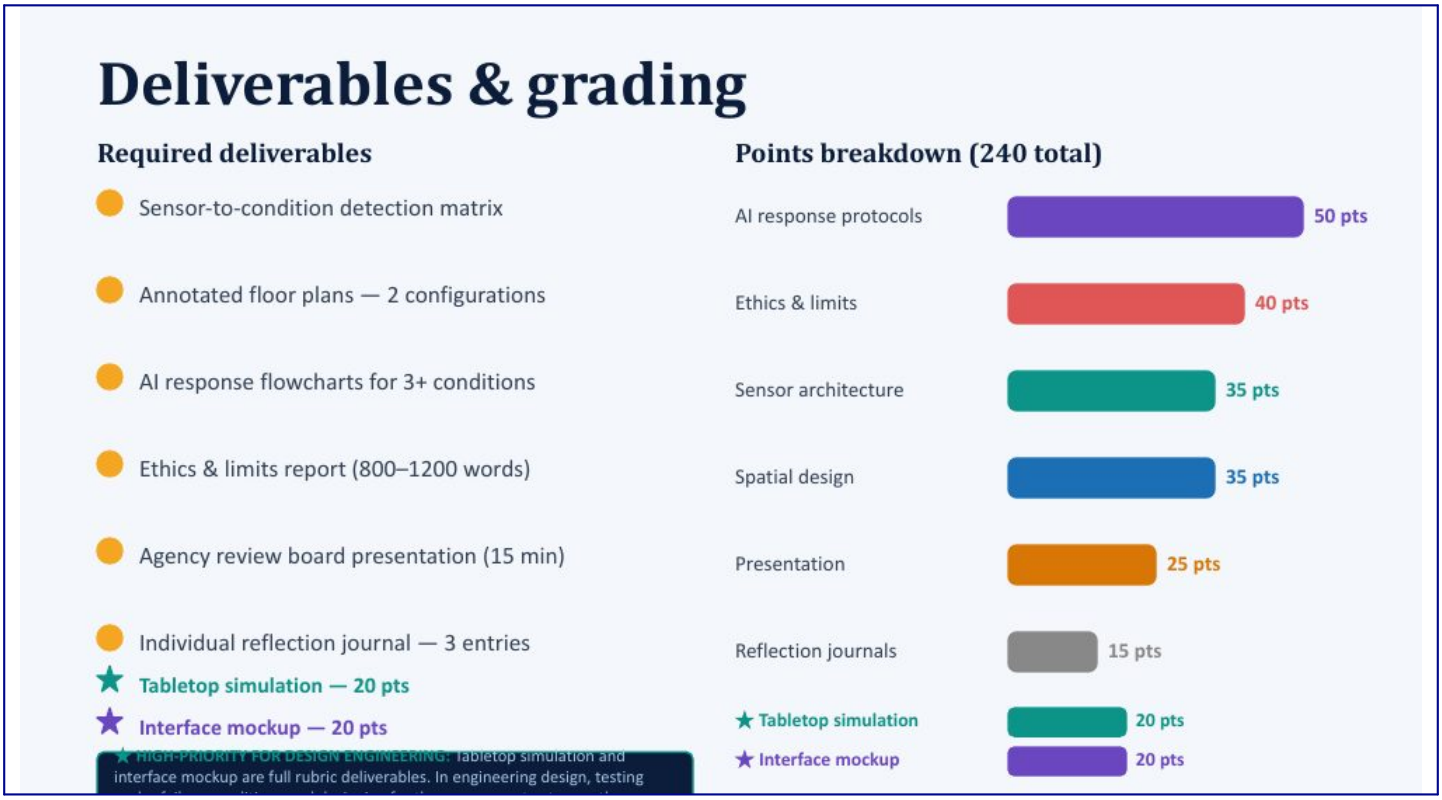
**Pictures from the simulations:**




*Realistic game*



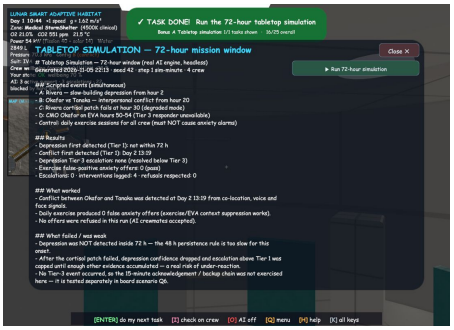

*Realistic game: night watch*




## ■ DONE ✓ WHAT WE DID WELL — Deliverables & grading (this deck: tabletop and interface 20 pts each)

- Tabletop simulation and crew interface are both complete in both games (this deck weights them 20 pts each — full marks claimed).

**Pictures from the simulations:**




*Realistic game*

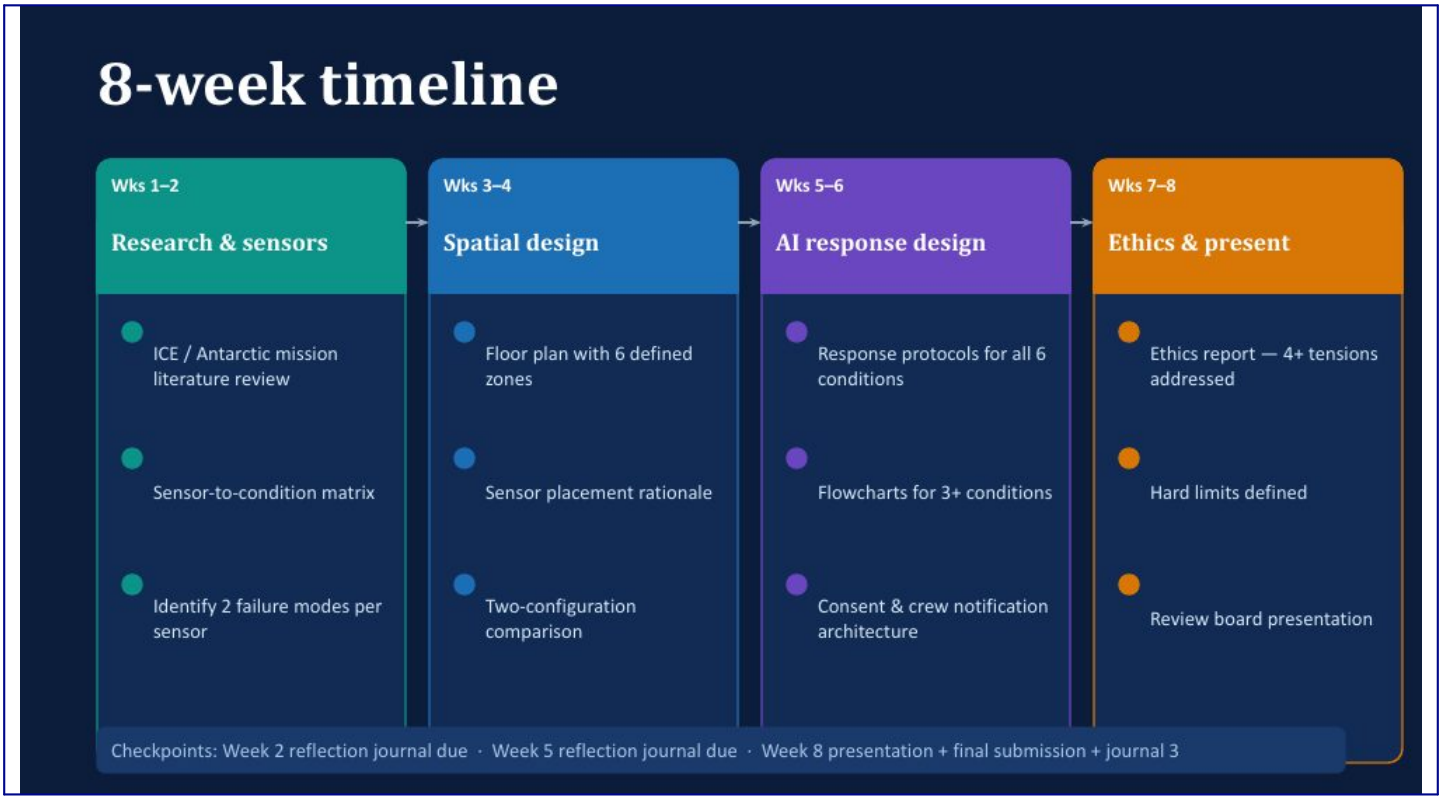



*Realistic game*

**Self-grade on this deck's scale (240):**

| Rubric line | Points | Our grade | What we did / why points were taken off |
|---|---|---|---|
| Sensor architecture and detection logic | +35 | **33/35** | Nine sensors, thresholds, 2+ failure modes each, mass/power, graceful degradation, live check-in. −2: thresholds are design values, not validated data. |
| Habitat spatial design | +35 | **30/35** | Six zones with purpose in both bases, sensor placement, Config A/B in the world. −5: no scaled, dimensioned drawing inside the games; crew of 4 instead of the brief's 6. |

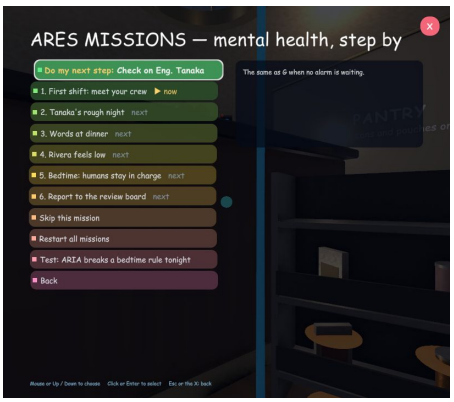
| Rubric line | Points | Our grade | What we did / why points were taken off |
|---|---|---|---|
| AI response protocol design | +50 | **46/50** | All six conditions × nine stages, live engine, tiers, consent, monitoring windows, fast CMO escalation. −4: flowcharts are text pages in the games (draw them on slides). |
| Ethics and limits analysis | +40 | **38/40** | Four tensions with design answers, hard limits enforced in code, override, night watch. −2: report must still be checked at 800–1,200 words when exported. |
| Presentation and communication | +25 | **22/25** | Run-of-show and Q1–Q6 live in the simulator. −3: delivery, teamwork and Q&A are judged on the day. |
| Individual reflection journals | +15 | **0/15** | Must be written by each student. Prompts are in the game. |
| Bonus A: tabletop simulation | +20 | **20/20** | 72-hour run with two simultaneous events + sensor failure + CMO on EVA; report saved. |
| Bonus B: crew interface mockup | +20 | **20/20** | Wellness, what the AI did, override, privacy, opt-out, preferences. |
| **Total (200 + 40 bonus)** | **240** | **209/240** | **87% of everything; 93% of the points the games themselves can earn (journals are written by each student).** |



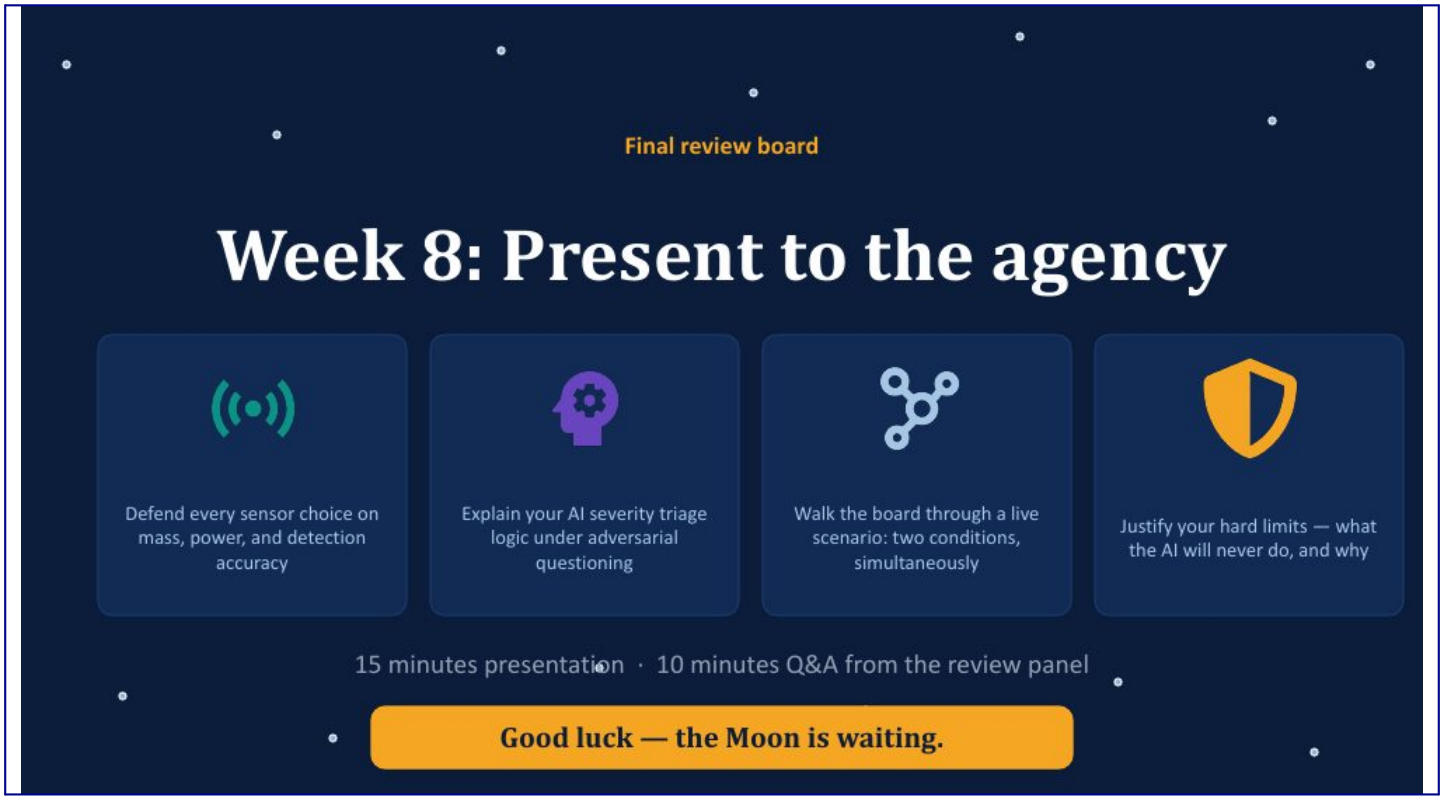

## ■ DONE ✓ WHAT WE DID WELL — 8-week timeline

- Each phase's output can be opened in the game in the order of the timeline (missions follow the same order).

**Pictures from the simulations:**




*Sci-fi game*



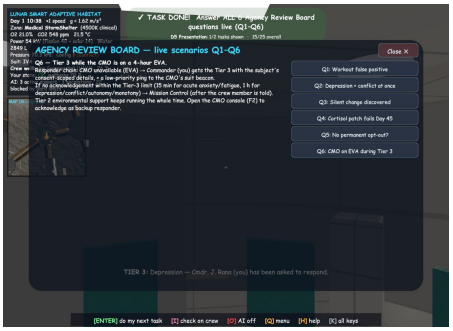

## ■ DONE ✓ WHAT WE DID WELL — Final review board

- All six board questions run live in both games.

**Pictures from the simulations:**




*Realistic game: Q1*




*Realistic game: Q6*